

RESEARCH INTERESTS

Statistics, Machine Learning, Generative Models, Computer Vision

EDUCATION

School of Mathematics (Zhuhai), Sun Yat-sen University

Sep 2022 – Jun 2026 (Expected)

- B.S. in Information and Computing Science, GPA: 4.00/4.00 (1/62)

Core curriculum: Mathematical Statistics, Real Analysis, Probability Theory, Numerical Analysis, Information Theory

ACADEMIC EXPERIENCE

Research Assistant, Human Cyber Physical Intelligence Laboratory, Sun Yat-sen University

Feb 2024 – Present

Advisor: Prof. Pengxu Wei, Prof. Liang Lin (Fellow of IEEE/IAPR)

- Research in machine learning and computer vision.
- Developed Minimum Variance Unbiased Autoencoder (MVU-AE) with a more effective latent representation for image reconstruction and generation tasks.
- First author of a paper is submitted to **ICML 2025**.

PUBLICATIONS

MVU-AE: Minimum Variance Unbiased Autoencoders

Under review at **ICML 2025**

Yaqin Chen, Junrong Lian, Qing Zhao, Xiangyang Ji, Liang Lin, Pengxu Wei

- Conducted a systematic analysis of two fundamental issues in VAEs: latent alignment bias and structural inductive bias, and identified their impact on generative modeling and representation learning.
- Proposed and formalized *deep* Minimum Variance Unbiased (*d*-MVU) conditions, inspired by MVUE in classical statistics, to improve the statistical efficiency of latent representations.
- Devised a novel autoencoder architecture that ensures compliance with *d*-MVU conditions, enhancing representation learning, generative quality, and latent space smoothness.

SELECTED PROJECTS

“Development of a Martial Arts Teaching and Scoring System Based on YOLOv8”

Dec 2023 – Dec 2024

2024 Chinese Academy of Sciences Innovation and Practice Training Program For Undergraduate

Project Leader

- Designed and implemented a real-time martial arts teaching and grading system leveraging YOLOv8 for advanced computer vision tasks.
- Improved martial arts training with interactive features, video analysis, and automated scoring mechanisms.
- Integrated a Large Language Model (LLM) to provide text and voice feedback, offering personalized suggestions and evaluations for improved learning outcomes.

AWARDS

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| • China National Scholarship (Top 1%, 1/151) | 2024 |
| • First-Class Excellent Student of SYSU (Top 5%, 1/151) | 2024 |
| • First-Class Excellent Student of SYSU (Top 5%, 1/151) | 2023 |
| • First Prize 37th Chinese Physics Olympiad of Jiangxi Province (Top 5%) | 2020 |

PROFESSIONAL SKILLS

- Programming Languages: Python, C++, MATLAB
- Languages: English(fluent), Chinese(native)

STANDARDIZED TESTS

- GRE: Verbal 129 (80%) +Quantitative 170 (92%) +Analytical Writing 3
- TOEFL: Total 100 (Reading 24+Listening 25+Speaking 25+Writing 26)